



Project: Ecommerce Customer Behavior

kaggle

Ecommerce Customer Behavior Dataset

Key Categories:

1. Purchase Behavior
2. Engagement
3. Customer Demographic
4. Customer Service
5. Financial Aspect

50 000 customers - 25 feat.

Age	float64
Gender	object
Country	object
City	object
Membership_Years	float64
Login_Frequency	float64
Session_Duration_Avg	float64
Pages_Per_Session	float64
Cart_Abandonment_Rate	float64
Wishlist_Items	float64
Total_Purchases	float64
Average_Order_Value	float64
Days_Since_Last_Purchase	float64
Discount_Usage_Rate	float64
Returns_Rate	float64
Email_Open_Rate	float64
Customer_Service_Calls	float64
Product_Reviews_Written	float64
Social_Media_Engagement_Score	float64
Mobile_App_Usage	float64
Payment_Method_Diversity	float64
Lifetime_Value	float64
Credit_Balance	float64
Churned	int64
Signup_Quarter	object



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Focus on 3 topics:

- Purchase Behavior (Q1 – Q4)
- Engagement (Q5 – Q9)
- Customer Demographic (Q10)
- Customer Service
- Financial & Status

What influences the Revenue ?



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1 – Purchase Behavior

- Check Customer Lifetime Value (CLV) distribution

SQL query

Seaborn plot

```
1 # Question 1:
2 # SQL query
3 query = "SELECT Lifetime_Value FROM data"
4
5 # use pandas to read the query and save into a table
6 table = pd.read_sql_query(query,database)
7 table.describe()
```

✓ 0.0s

	Lifetime_Value
count	50000.000000
mean	1440.626292

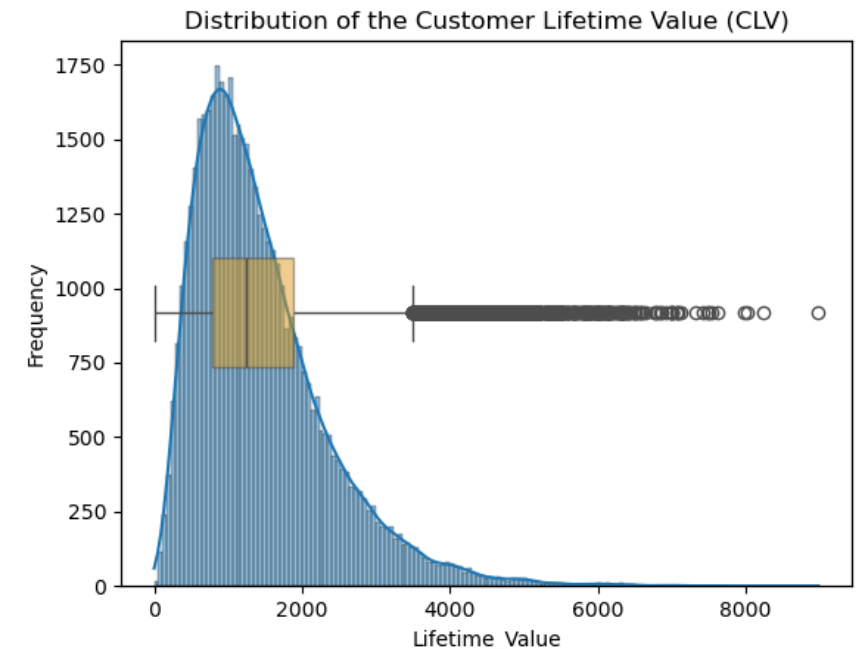
Need to import sqlite3 to run it !

```
1 fig, ax = plt.subplots()
2
3 # Histogram
4 figure = sns.histplot(table["Lifetime_Value"], kde=True, ax=ax)
5 plt.ylabel('Frequency')
6
7 # Create second axis sharing x-axis
8 ax2 = ax.twinx()
9
10 # Boxplot (horizontal, placed above along y)
11 sns.boxplot(
12     x=table["Lifetime_Value"],
13     ax=ax2,
14     width=0.2,
15     color = 'orange',
16     boxprops=dict(alpha=0.5), # transparency
17 )
18
19 # Hide second y axis
20 ax2.set_yticks([])
21 ax2.set_ylabel("")
22
23 # Title
24 plt.title("Distribution of the Customer Lifetime Value (CLV)")
25
26 # Save as png
27 figure.figure.savefig('CLV_distribution.png')
28 plt.show()
```

What is customer lifetime value?

Customer Lifetime Value (CLV or LTV) is a metric that predicts the net profit your brand will realize over the entire lifespan of that specific customer relationship. It measures how much revenue you can expect one customer to generate over the course of your business relationship.

www.pegas.com





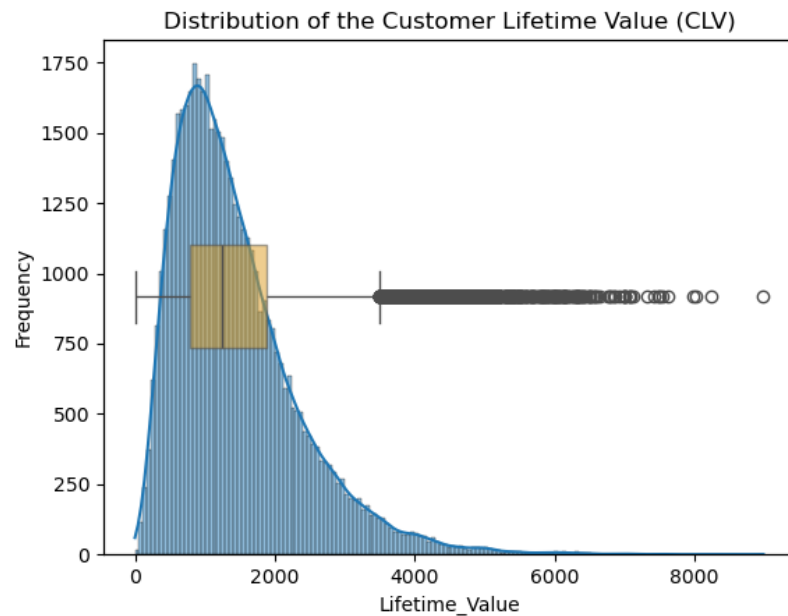
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1 – Purchase Behavior

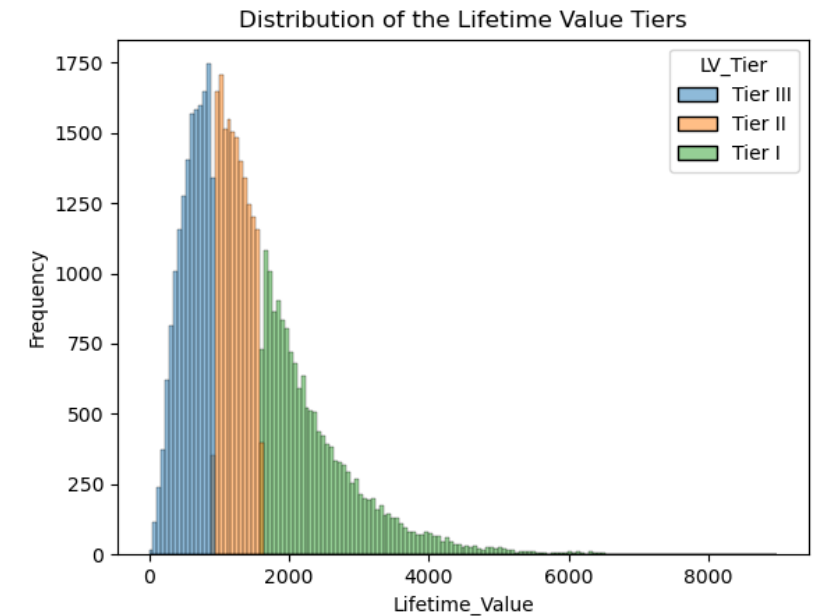
- Check Customer Lifetime Value (CLV) distribution
- Segmentation of the customers into 3 Tiers

Segmentation



```
1 # Question 2: Customer Segmentation
2 # Set the thresholds to 33% and 66%
3 q1 = table["Lifetime_Value"].quantile(0.33)
4 q2 = table["Lifetime_Value"].quantile(0.66)
5
6 # Create a function to run on the LV column
7 def segment(lifetime):
8     # Assign Tier III for lowest LVs
9     if lifetime < q1:
10         return "Tier III"
11     # Assign Tier II for medium LVs
12     elif lifetime < q2:
13         return "Tier II"
14     # Assign Tier I for highest LVs
15     else:
16         return "Tier I"
17
18 # Apply the function to the column
19 table["LV_Tier"] = table["Lifetime_Value"].apply(segment)
20 table
```

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1 – Purchase Behavior

- How much do each group participate in the total CLV ?

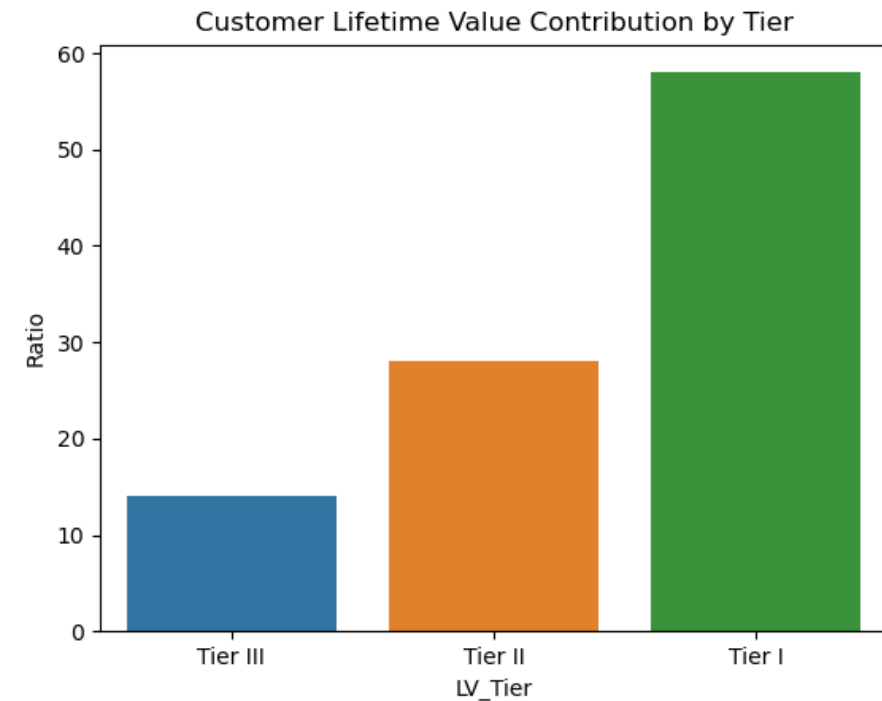
```
1 # SQL query
2 query = "SELECT LV_Tier, COUNT(LV_Tier) FROM data_with_LV_tier GROUP BY LV_Tier"
3
4 # use pandas to read the query and save into a table
5 table = pd.read_sql_query(query,database)
6 table
```

✓ 0.0s

	LV_Tier	COUNT(LV_Tier)
0	Tier I	17000
1	Tier II	16501
2	Tier III	16499

➤ 3 Tiers have same population size ($\pm 3\%$)

➤ **Tier 1 represents 58% of the CLV !**



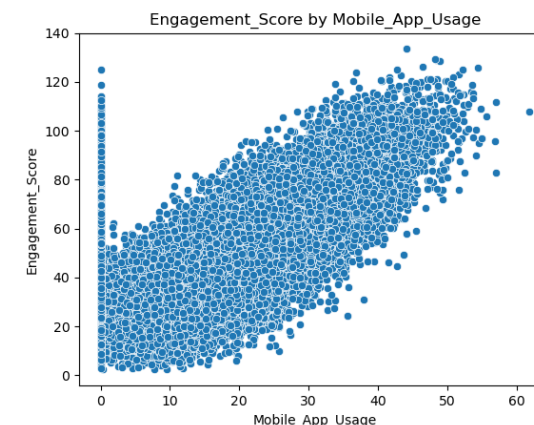
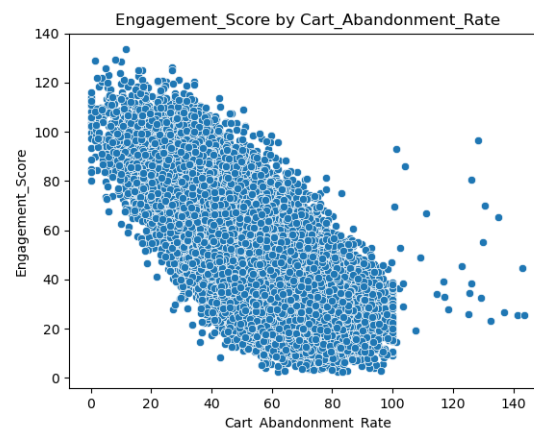
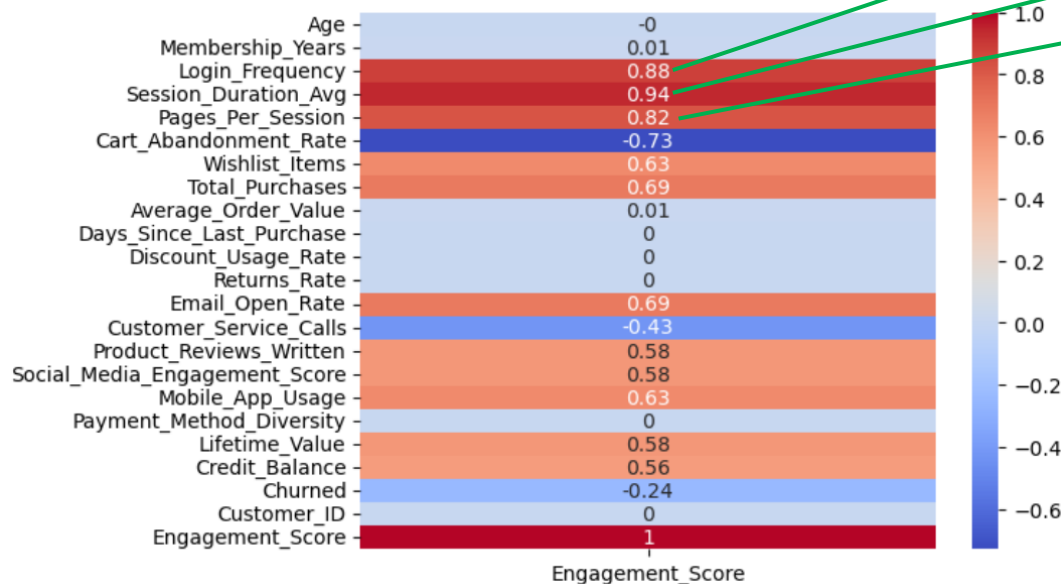


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2 – Engagement

- Set a definition (arbitrary):
- Which factors influence engagement ?

```
1 # Question 5
2 # Defining Engagement:
3 # Engagement_Score = Login_Frequency + Session_Duration_Avg + Pages_Per_Session
4
5 df["Engagement_Score"] = (df["Login_Frequency"] + df["Session_Duration_Avg"] + df["Pages_Per_Session"])
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```



- Cart Abandonment = **Negative corr.**
- Mobile App = **Positive corr.**



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3 – Customer Demographic

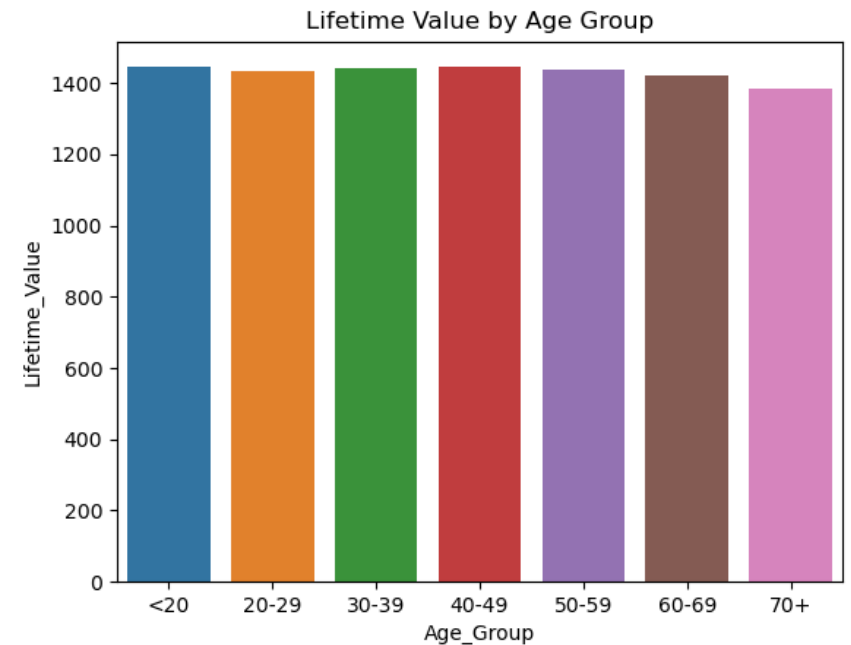
- Gender vs CLV

```
1 query = "SELECT Gender,AVG(Lifetime_Value) as avg_value " \
2 "FROM data_with_LV_Tier " \
3 "GROUP BY Gender;"
4 table = pd.read_sql_query(query,database)
5 table
```

0.0s

Gender	avg_value
Female	1442.153550
Male	1438.803539
Other	1446.272871

- Age Group vs CLV



➤ Customer Demographic (gender, age) reveal a negligible influence on the CLV.



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Conclusions: **What influences the CLV/revenue ?**

- **Purchase Behavior:** focus on Tier I customers (58% of CLV)
- **Engagement:** positively influence CLV (keep pushing Mobile App., webshop design, Social Media presence)
- **Customer Demographic:** failed to show any influence on CLV
- To go further:
 - Investigate **Financial** and **Customer Service** topics
 - Improve **Customer Demographics** (collect new data, i.e. education, family status ...)



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Challenges :

Problems

- Many to many join problem
- Add new column to database to use in SQL
- Data cleaning
 - > (Ages > 110, purchases < 0)
- Plot customization (size, fitting, overlap, subplots)

Solutions

- Use index to merge (not feature)
- `pd.to_sql` function
- `df.loc` (find indexes) of outliers
 - > Set to median



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Major obstacle :

Keep the data clean !

Merge / join alter size of data

NaN (0 ? Median ?)

Wrong input (Age, Total_Purchase)

Engagement Score

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Thank You !